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United States Patent	12404016
Kind Code	B2
Date of Patent	September 02, 2025
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Aircraft and power device

Abstract

To provide an aircraft having improved efficiency and flexibility of the airframe configuration when installing a functional part in the airframe, and a power unit having a configuration for the improved efficiency and flexibility. To provide an aircraft, comprising: a power unit having a first space open at least one of its upper or lower surface; a propeller having a first through-space and connected to the power unit; and a functional part having a predetermined function and located at least partially in an internal space formed by the first space and the first through-space.

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Appl. No.:	18/004439
Filed (or PCT Filed):	July 07, 2020
PCT No.:	PCT/JP2020/026607
PCT Pub. No.:	WO2022/009319
PCT Pub. Date:	January 13, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240025574 A1	Jan. 25, 2024

Publication Classification

Int. Cl.: B64U30/299 (20230101); B64U10/16 (20230101); B64U20/94 (20230101); B64U50/19 (20230101); H02K7/00 (20060101); H02K9/22 (20060101)

U.S. Cl.:

CPC B64U20/94 (20230101); B64U10/16 (20230101); B64U30/299 (20230101); B64U50/19 (20230101); H02K7/006 (20130101); H02K9/227 (20210101)

Field of Classification Search

CPC: B64U (50/19); B64U (60/50); B64U (60/70); B64D (45/04); B64D (45/06)

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Background/Summary

TECHNICAL FIELD

(1) This invention relates to an aircraft and power device.

BACKGROUND ART

(2) In recent years, the use of drones and unmanned aerial vehicles (UAVs) and other flying vehicles (hereinafter collectively referred to as “aircraft”) has been increasing. In line with this trend, efforts are underway to improve the performance of aircrafts and to develop new services that use these aircrafts. In practice, various parts may be attached to its airframe to add predetermined functions to the aircraft, and Patent Literature 1 discloses an aircraft equipped with landing legs that are capable of cushioning the impact of landing.

PRIOR ART LIST

Patent Literature

(3) [Patent Literature 1] WO2016/179827

SUMMARY OF THE INVENTION

Technical Problem

(4) However, the Aircraft of Patent Literature 1 uses a method of directly attaching the landing leg support members to an arm or frame of the airframe (hereinafter collectively referred to as the “holding part”), and it is preferable to have fewer components from the viewpoint of improving the efficiency of the airframe configuration of the aircraft. It is desirable to be able to provide parts having a predetermined function or parts supporting parts (hereinafter referred to as “functional parts”) without significantly changing the conventional airframe configuration.

(5) In addition, since the propeller of the Patent Literature 1 is a traction type (pull type), the motor and propeller (hereinafter collectively referred to as the “rotor blade part”) are attached to the upper part of the holding part, and the functional part (landing legs) can be attached to the bottom of the holding part. However, when the propeller is a propulsion type (pusher type), the rotor blades are already attached to the lower part of the holding part, and if the functional part is to be attached to the lower part of the holding part, the placement of the propeller is limited by the rotor blades. In particular, in the case of landing legs, it is difficult to create a highly stable arrangement during landing (e.g., near the rotor blades, which are located away from the center of the fuselage). Therefore, especially in the case of landing legs, the configuration should be usable for both traction (pull type) and propelled (pusher type) types of aircraft. The same is true for other functional parts in that their placement is also restricted.

(6) Therefore, one object of the present invention is to provide an aircraft with improved efficiency and flexibility of the airframe configuration when providing functional parts on the airframe, and a power unit with a configuration for improved efficiency and flexibility.

Technical Solution

(7) According to the present invention, it is possible to provide an aircraft comprising: a power unit having a first space open at least one of its upper or lower surface; a propeller having a first through-space and connected to the power unit; and a functional part having a predetermined function and located at least partially within an internal space formed by the first space and the first-through space.

(8) According to the present invention, it is possible to provide a power unit, wherein the power unit has a first space open at least one of its upper or lower surface, wherein the power unit comprises a functional part providing a predetermined function, wherein the functional part is at least partially located in the first space, and a configuration that does not rotate.

(9) According to the present invention, it is possible to provide a power unit, wherein the power unit has a first space with at least one of its upper or lower surfaces open. wherein the power unit

has a support part connected to a function-providing part that provides a predetermined function, wherein the support part is at least partially located in the first space and includes a non-rotating configuration.

Advantageous Effects

(10) According to the present invention, an aircraft that improves the efficiency and flexibility of the aircraft configuration, and a power unit with a configuration for improving the efficiency and flexibility can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a top conceptual view of a power unit of an embodiment of this invention.
- (2) FIG. 2 is a conceptual view of the power unit of FIG. 1 from the side with the propeller connected.
- (3) FIG. 3 shows a conceptual view of the power unit of FIG. 2, disassembled and viewed from the side.
- (4) FIG. 4 shows a side view of the aircraft in the first embodiment of the invention.
- (5) FIG. 5 shows a side view of the aircraft in the second and third embodiments of the invention.
- (6) FIG. 6 shows a conceptual view of the power unit in other embodiment of the invention, disassembled and viewed from the side.
- (7) FIG. 7 shows a functional block diagram of an aircraft in an embodiment of the present invention.
- (8) FIG. 8 shows a side view of the aircraft in the fourth embodiment of the present invention.
- (9) FIG. 9 shows a top view of the aircraft of FIG. 8.
- (10) FIG. 10 shows a side view of an aircraft in the existing technology when it is equipped with a propeller guard.
- (11) FIG. 11 shows a top view of the aircraft of FIG. 10.
- (12) FIG. 12 shows a side view of the aircraft in the fifth embodiment of the invention.
- (13) FIG. 13 shows a side view of an aircraft in the existing technology when it is equipped with a nozzle.
- (14) FIG. 14 shows a side view of an aircraft in a sixth embodiment of the invention.
- (15) FIG. 15 shows a side view of an aircraft in a seventh embodiment of the invention.
- (16) FIG. 16 shows a side view of an aircraft in the seventh embodiment of the invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (17) The following is a list and description of the contents of the embodiments of the present invention. The aircraft according to the embodiment of the invention consists of the following:
- (18) [Item 1]
- (19) An aircraft, comprising: a power unit having a first space open at least one of a upper surface or a lower surface; a propeller having a first through-space and connected to the power unit; and a functional part having a predetermined function and located at least in part within the internal space formed by the first space and the first through-space.
- [Item 2]
- (20) The aircraft according to Item 1, wherein the functional part includes a non-rotating configuration.
- [Item 3]
- (21) The aircraft of Item 1 or Item 2, wherein a portion of the functional part protrudes from the first through-space of the propeller.
- [Item 4]
- (22) The aircraft of any one of Items 1 or 3, wherein the first space is a second through-space

through the power unit.

[Item 5]

(23) The aircraft of any one of Items 1 to 4, wherein the functional part includes a grounding part that is in contact with the ground when landed.

[Item 6]

(24) The aircraft of any one of Items 1 to 5, wherein a portion of the functional part protrudes from the other side of the power unit, which is different from the one side of the power unit in contact with the propeller.

[Item 7]

(25) The aircraft of any one of Items 1 to 6, wherein the functional part comprises: a support part at least partially located in the internal space; and a function-providing part connected to the support part and providing the predetermined function.

[Item 8]

(26) The aircraft according to Item 7, wherein the aircraft includes a grounding part that is in contact with the ground at the time of landing as the function-providing part.

[Item 9]

(27) The aircraft of Item 7 or Item 8, wherein the aircraft includes a propeller guard as the function-providing part.

[Item 10]

(28) The aircraft of any one of Item 7 to Item 9. wherein the function-providing part includes a jetting means.

[Item 11]

(29) A power unit, wherein the power unit has a first space open at least one of a upper surface or a lower surface, wherein the power unit is provided with a function part providing a predetermined function, and wherein the functional unit is at least partially located in the first space and includes a non-rotating configuration.

[Item 12]

(30) A power unit, wherein the power unit has a first space open at least one of a upper surface or a lower surface, wherein the power unit has a support pan connected to a function-providing part that provides a predetermined function, and wherein the support part is at least partially located in the first space and includes a non-rotating configuration.

[Item 13]

(31) The power unit of Item 11 or Item 12, wherein the first space is a through-space through the power unit.

Details of Embodiments According to this Invention

(32) An aircraft according to the embodiments of the present invention will be described below with reference to the drawings. In the accompanying drawings, identical or similar elements are given identical or similar reference codes and names, and duplicate descriptions of identical or similar elements may be omitted in the description of each embodiment. The features shown in each embodiment are also applicable to other embodiments as long as they do not contradict each other.

(33) FIGS. **1** and **2** illustrate an outer rotor type motor as a power unit **20** provided by an aircraft according to an embodiment of the invention. The power unit **20** has, for example, a rotor **22** with a permanent magnet **21** and a stator **24** with a winding **23**. The propeller **110** is connected to the power unit **20** and rotates in response to the rotation of the power unit **20** to generate lift from a lift-generating surface of the propeller **110**.

(34) The configuration of the power unit **20** is not limited to this, as long as it has an inner space and a functional part **10** described below. For example, in the case of an inner rotor type motor, the configuration may be a motor having a pipe-like hollow structure and a hollow shaft, with the through space in the hollow shaft used as a first space described below.

(35) The power unit **20** has a space **25** (hereinafter referred to as the “first space **25**”) located in the center of the top view, for example, as illustrated in FIG. **1** and FIG. **3**, etc. This first space **25** is open at least one of the upper or lower surface of the power unit **20**, allowing the functional part **10** to protrude from the power unit **20**. For example, in FIG. **3**, only the upper surface of the power unit **20** is open, and in FIG. **4**, the upper and lower surfaces of the power unit **20** are open. In FIG. **1**, the opening of the first space **25** is located in the center of the top view, but it is not limited to the center of the top view, and may be located at any location as long as the functional part **10** does not interfere with the operation of the propeller **110**. The propeller **110** also has a first through-space **112**. The space in which the first space **25** of the power unit **20** and the first through-space **112** of the propeller **110** are connected to each other is referred to as the “internal space”. Each of these spaces may be straight, bent, or curved, respectively, and their shape is not particularly limited.

(36) The functional part **10** provided by the aircraft according to the present invention may be configured to protrude from the top view center of the stator **24** and to project the rotor **22** from the power unit **20** through the first space **25**, as shown, for example, in FIG. **3**. Furthermore, the functional part **10** is also configured to protrude through the propeller **110**, so that the functional part **10** itself functions as, for example, a landing leg of the aircraft **100**, as shown, for example, in FIG. **4**. Also, as shown in FIG. **5**, for example, the functional part **10** may comprise a support **11** at least partially located in the internal space and the function-providing part **12** connected to the support **11** and providing a predetermined function (e.g., landing legs).

(37) FIG. **6** illustrates a modification of the functional part **10** provided by the aircraft of an embodiment of the present invention. The functional part **10** may protrude from a configuration external to the power unit **20**, such as the aircraft's holding part **120a** and **120b** (including, for example, a motor mount (not shown), if the aircraft has one) that holds the power unit **20** and a rotor blade part including the propeller **110**. The functional part **10** may also protrude indirectly through a member (not shown) that secures the functional part **10**. These functional part **10** can then be configured to protrude from the power unit **20** and the propeller **110** through the first space of the power unit **20** and the first through-space of the propeller **110**.

(38) The functional part **10** does not touch the rotor **22** of the power unit **20**, and at least the support **11** may be configured not to rotate, or the entire functional part **10** may be configured not to rotate. Alternatively, if the power unit **20** has a through shaft, the functional part **10** may be configured so that the power unit **20** does not touch the through shaft, or may be configured to contact the through shaft using bearings or the like. By doing so, the functional part **10** may be configured not to rotate without being affected by the rotation of the power unit **20**. These are only simplified examples, and necessary components may be added as necessary.

(39) Thus, by utilizing the internal spaces of the power unit **20** and the propeller **110**, the functional part **10** with a predetermined function can be easily provided without major changes to the conventional configuration of the airframe. In addition, regardless of whether the propeller **110** is a traction type (pull type) or a propulsion type (pusher type), the functional part **10** can be arranged using the spaces above and below the rotor blades, so the arrangement is not limited in any particular way.

(40) The functional part **10** can have a variety of functions. The functional part **10** can be, for example, a landing leg, lights, an antenna, a propeller guard, jetting means (a nozzle), a heat sink, a rectification device, a wheel, an aerodynamic part, and components for supporting the onboard objects.

(41) The function provider **12** may be configured as multiple types of attachments that can be connected to and interchangeable with the support **11** according to requirements and other factors. In the case of interchangeability, the attachment of the support **11** should be standardized so that multiple types of attachments can be easily interchanged. The attachment connection configuration of the attachment part can be easily interchangeable by providing well-known connection methods such as connectors and screws.

(42) Since the functional part **10** can operate independently of the rotation of the power unit **20**, a servo or motor installed separately from the power unit **20** can be used to perform the specified rotation or oscillation. This changes, for example, the direction of the nozzle or the angle of the aerodynamic part.

(43) <Configuration of Aircraft **100**>

(44) The aircraft **100** illustrated in FIG. **4** and elsewhere should be equipped with at least the propeller **110**, power unit **20**, and other elements for flight, and energy (e.g., secondary batteries, fuel cells, fossil fuel, etc.) to operate them.

(45) The aircraft **100** shown in the figure is depicted in simplified form to facilitate the explanation of the structure of the invention, and the detailed configuration of the control unit, for example, is not shown.

(46) The aircraft **100** may, for example, travel in the direction of arrow D (−Y direction) in the figure. In the following description, the terms may be used according to the following definitions: Front-back direction: +Y direction and −Y direction, up-down direction (or vertical direction): +Z direction and −Z direction, left-right direction (or horizontal direction): +X and −X directions, forward direction (forward): −Y direction, backward direction (backward): +Y direction, ascending direction (upward): +Z direction, descending direction (downward): −Z direction.

(47) Propellers **110a** and **110b** rotate under the output from the power unit **20**. The rotation of the propellers **110a** and **110b** generates propulsive force to take the aircraft **100** off from its starting point, move it, and land it at its destination. The propellers **110a** and **110b** can rotate to the right, stop, and rotate to the left, etc.

(48) The propeller **110** provided by the aircraft **100** of this invention has one or more blades. Any number of blades (rotors)(e.g., 1, 2, 3, 4, or more blades) is acceptable. The shape of the blades can be any shape, such as flat, curved, kinked, tapered, or a combination thereof. The shape of the blades can be changeable (e.g., stretched, folded, bent, etc.). The blades can be symmetrical (having identical upper and lower surfaces) or asymmetrical (having differently shaped upper and lower surfaces). The blades can be formed into airfoils, wings, or any geometry suitable for generating dynamic aerodynamic forces (e.g., lift, thrust) when the blades are moved through the air. The geometry of the vane can be selected as appropriate to optimize the dynamic aerodynamic characteristics of the vane, such as increasing lift and thrust and reducing drag.

(49) The propeller provided by the aircraft **100** may be, but is not limited to, fixed pitch, variable pitch, or a mixture of fixed and variable pitch.

(50) The power unit **20** produces the rotation of the propeller **110**. For example, the drive unit can include an electric motor or an engine. The blades can be driven by the motor and rotate around the motor's axis of rotation (e.g., the long axis of the motor).

(51) The blades can all rotate in the same direction or can rotate independently. Some of the blades rotate in one direction while others rotate in the other direction. The blades can all rotate at the same RPM, or they can each rotate at a different RPM. The number of rotations can be determined automatically or manually based on the dimensions of the moving object (e.g., size, weight) and control conditions (speed, direction of movement, etc.).

(52) The aircraft **100** determines the number of revolutions of each motor and the angle of flight according to the wind speed and direction by means of a flight controller or a propo, or the like. This allows the aircraft to move up and down, accelerate and decelerate, and change direction.

(53) The aircraft **100** can fly autonomously according to routes and rules set in advance or during the flight, or by using a propo to control the aircraft.

(54) <Functional Block Diagram of Aircraft **100**>

(55) FIG. **7** shows a functional block diagram of the aircraft **100**, and the functional blocks in FIG. **7** are a minimum reference configuration. A flight controller is a so-called processing unit. A processing unit can have one or more processors, such as a programmable processor (e.g., central processing unit (CPU)). The processing unit has a memory, not shown, which is accessible. The

memory stores logic, code, and/or program instructions that can be executed by the processing unit to perform one or more steps. The memory may include, for example, a separable medium such as an SD card, random access memory (RAM), or an external storage device. Data acquired from a camera and sensors may be directly transmitted to and stored in the memory. For example, still and moving image data captured by a camera or other device is recorded in the internal or external memory.

(56) The processing unit includes a control module configured to control the state of the rotorcraft. For example, the control module controls the propulsion mechanism of the rotorcraft (e.g., the 20 power unit) to adjust the spatial placement, velocity, and/or acceleration of the rotorcraft having six degrees of freedom (translational motion x , y , and z , and rotational motion θ_x , θ_y , and θ_z). The control module can control one or more of the states of the mounting part and sensors.

(57) The processing unit is capable of communicating with a transmission/reception unit configured to transmit and/or receive data from one or more external devices (e.g., terminals, display units, or other remote controllers). The transmission/reception unit can use any suitable means of communication, such as wired or wireless communication. For example, the transmission/reception unit can use one or more of the following: local area network (LAN); wide area network (WAN); infrared, wireless; WiFi, point-to-point (P2P) network; telecommunications network; or cloud communications. The transmission/reception unit can transmit and/or receive one or more of the following: data acquired by sensors, processing results generated by the processing unit, predetermined control data, and user commands from a terminal or remote controller.

(58) Sensors in this embodiment can include inertial sensors (accelerometers, gyro sensors), a GPS sensor, a proximity sensor (e.g., lidar), or a vision/image sensor (e.g., a camera).

Details of the First Embodiment

(59) FIG. 4 shows an example of the first configuration of the invention, in which the functional part **10** functions as a landing leg.

(60) As mentioned above, in an aircraft **100** performing vertical takeoff and landing, when the aircraft **100** contacts the landing surface, the wider the spacing between the landing legs, the more stable it will be. Then, the location of the landing legs where the spacing can be widened without extending the holding part **120**, for example, would be near the bottom of the rotor blades. However, the landing legs, especially when the propeller **110** is a propulsion type, need to be installed avoiding the propeller rotating surface so that they do not come into contact with the propeller **110**. Therefore, the spacing between landing legs becomes narrower when they are avoided on the center side of the fuselage, and when they are avoided on the outside side of the fuselage, the holding part **120** needs to be extended for landing leg connection. Therefore, with the conventional configuration, it is difficult to achieve both improved landing stability and improved aircraft efficiency, especially when the propeller **110** is a propulsion type.

(61) Therefore, in the first embodiment shown in FIG. 4, the functional part **10** uses the internal space comprising the first space **25** of the power unit **20** and the first through-space **112** of the propeller **110** to allow a landing leg below the power unit **20** even in the aircraft **100** whose propeller **110** is propulsive.

(62) The landing legs may have a ground contact part that makes contact with the ground and may also be equipped with dampers or the like that provide shock mitigation during landing or when placing the aircraft.

(63) This makes it possible to both improve landing stability and aircraft's fuselage efficiency, even when the propeller **110** is a propulsion type.

(64) In the design of aircrafts, the power unit **20** connected to the propeller **110** and the holding part **120a** and **120b** are generally mounted robustly in terms of the assumed loads. Therefore, especially when the functional part **10** is a landing leg, it is possible to install the landing leg, which is subject to loads during landing, in the same or close location as the power unit **20**, etc., so that the locations to be made robust can be centralized, thus reducing weight increase and center of gravity

dispersion.

Details of the Second Embodiment

(65) FIG. 5 illustrates the second embodiment of the invention, in which the function-providing part **12** of the functional part **10** functions as a landing leg.

(66) As illustrated in FIG. 5, for example, in aircraft **100** using a traction-type propeller **110**, the landing legs are not restricted in their placement on the propeller **110** when they are provided near the lower portion of the rotor blades and holding pan **120a** and **120b**, as in the conventional case. However, by using the internal space comprised by the first space **25** of the power unit **20** and the first through-space **112** of the propeller **110**, it is possible to provide the landing legs with a simpler configuration. As illustrated in FIG. 5, the functional part **10** may comprise the support **11** protruding from the power unit **20** and the function-providing part **12** connected to the support **11** and providing a function as a landing leg.

Details of the Third Embodiment

(67) FIG. 5 illustrates a third embodiment of the invention in which the function-providing part **12** of the functional part **10** functions as a lighting device (e.g., LED, etc.) or an antenna device.

(68) As illustrated in FIG. 5, it is also possible to provide the functional pan **10** above the rotor blade section, even for the aircraft **100** that uses, for example, a traction-type propeller **110**. More specifically, for example, a lighting device or an antenna device can be provided above the rotor blade section as the function-providing part **12**. Although the configuration shown in FIG. 5 is for the aircraft **100** that uses a traction-type propeller **110**, it can alternatively be provided for the aircraft **100** that uses a propulsion-type propeller **110**. The functional pan **10** is not limited to a configuration provided on the connection side of the propeller **110**, but can be used in any configuration, such as above or below a propulsion type aircraft, above or below a traction type aircraft, or both above and below a propulsion type and traction type aircraft. This can improve the efficiency of aircrafts by reducing the increase in weight of the airframe by, for example, reducing the number of component parts, and by improving the flexibility of the spatial arrangement.

Details of the Fourth Embodiment

(69) FIGS. 8 and 9 illustrate a fourth embodiment of the invention in which the function part **20**, or the function providing-part **23** of the function part **20**, functions as a propeller guard.

(70) Since the propeller guard can be installed from the top of the aircraft **100**, the installation and removal of the propeller guard while the aircraft **100** is on the ground is easier than the installation from the side or from below, as shown in FIGS. 10 and 11.

(71) The configuration shown in FIGS. 8 and 9 is for the aircraft **100** that uses a traction-type propeller **110**, but it is also possible to provide the functional part **10** on the aircraft **100** that uses a propulsion-type propeller **110** instead. The functional pan **10** is not limited to a configuration provided on the connection side of the propeller **110**, but can be used in any configuration, such as above or below a propulsion-type aircraft, above or below a traction-type aircraft, or both above and below a propulsion-type and traction-type aircraft. This can improve the efficiency of aircrafts by reducing the increase in weight of the airframe by, for example, reducing the number of component parts, and by improving the flexibility of the spatial arrangement.

Details of the Fifth Embodiment

(72) FIG. 12 shows an example of the fifth embodiment of the invention, in which the functional part **10**, or the function-providing part **12** of the functional pan **10**, serves as the jetting method.

(73) Functional part **10** is a pipe or hose through which a predetermined liquid or powder is passed, and if necessary, as illustrated in FIG. 12, a pipe or the like may be part of the configuration of support part **11**, and a jetting means (nozzle) connected to the pipe or the like that sprays, atomizes, or jets the liquid or powder may be used as functional-providing part **12**. By setting the function-providing part **12** of the function part **10** on the lower side of the aircraft **100**, the propeller **110**'s wake can be used to strongly apply the liquid or powder to the object or to send it farther away.

(74) In existing multicopter pesticide spraying aircraft, when utilizing the same effect, a pipe or

other external device is provided separately from the aircraft's holding part **120a** and **120b** to pass liquid or powder particles, as shown in FIG. **13**. In the functional part **10** according to the present invention, by providing a pipe or the like for passing liquid or powder/pellets inside the holding part **120a** and **120b** of the aircraft, air resistance during flight can be reduced compared to the flying vehicle **100** that has a separate pipe or the like on the outside.

(75) The configuration shown in FIG. **12** is for the aircraft **100** that uses a traction-type propeller **110**, but it can alternatively be provided for the aircraft **100** that uses a propulsion-type propeller **110**. However, the configuration is not limited to installing the functional part **10** on the connection side of the propeller **110**, but can be used in any configuration, such as installing the functional part **10** above or below the propulsion-type aircraft, above or below the traction-type aircraft, or both above and below the propulsion-type and traction-type aircrafts. This makes it possible to improve the efficiency of the aircraft, such as by reducing the weight increase of the airframe by reducing the number of component parts and by improving the flexibility of the spatial arrangement. Furthermore, if the direction of the jet is the same as the direction of the propeller wake flow, as described above, it is expected to efficiently utilize the propeller wake flow, etc.

Details of the Sixth Embodiment

(76) FIG. **14** shows an example of the sixth embodiment of the invention, in which the function part **10**, or the function-providing part **12** of the function part **10**, functions as a heat sink.

(77) As illustrated in FIG. **14**, the function part **10**, or the function-providing part **12** of the function part **10**, can be provided as a heat sink for the power unit **20** by utilizing the wake motion generated by the rotation of the propeller **110**. The use of a material with high thermal conductivity not only draws heat generated inside the power unit **20** to the outside and increases the area exposed to the outside air, but also makes it possible to efficiently cool the power unit **20** because the airflow near the heat sink is difficult to stagnate due to the wake flow of the propeller **110**.

(78) In particular, a power unit **20** that is designed to be waterproof and drip-proof is highly sealed to prevent water and dust from entering the unit. In the power unit **20**, it is difficult to cool the inside of the power unit **20** because outside air is not drawn into the power unit **20**. Therefore, improving cooling performance by enlarging the heat sink is effective.

(79) Examples of heat sink configurations include a pipe shape called a heat pipe, which is used in well-known heat sink mechanisms, and a Sirocco fan shape that takes into consideration not to obstruct the airflow of the propeller wake.

Details of Seventh Embodiment

(80) FIGS. **15** and **16** illustrate the seventh embodiment of the invention, in which the function part **10**, or the function-providing part **12** of the function part **10**, functions as a rectifier.

(81) When the propeller **110** provided by the aircraft **100** rotates, a wake is generated. As illustrated in FIG. **15**, the function part **10**, or the function-providing part **12** of the function part **10**, is provided as an airflow rectifier on the side where the wake is generated, thereby preventing the generation of vortices in the wake that lead to reduced flight efficiency and enabling improved flight efficiency.

(82) As shown in FIG. **16**, if the aircraft **100** was a VTOL aircraft, the power unit **20** would be used in the forward and backward directions during horizontal flight, etc., except during vertical takeoff and landing and hovering. In this case, the aircraft **100**'s flight efficiency is improved by providing the functional part **10** or the functional-providing part **12** of the functional part **10** as an airflow rectifier on the propeller connection side and the opposite side of the propeller connection side of the power unit **20**, which are in front of and behind the power unit **20**.

(83) If the functional part **10**, or the function-providing part **12** of the functional part **10**, has multiple functions, for example, a rectifying shape may be provided below the fuselage using a metallic material that acts as a heat sink. In this case, the function part **10**, or the function-providing part **12** of the function part **10**, can serve as a heat sink, a rectifier, and landing legs. By consolidating the functions in this manner, it is possible to prevent an increase in aerodynamic drag

and weight.

(84) The above-described embodiments are merely examples to facilitate understanding of the invention and are not intended to be construed as limiting the invention. It goes without saying that the invention may be changed and improved without departing from its purpose, and that the invention includes its equivalents.

DESCRIPTION OF REFERENCE NUMERALS

(85) **10** Functional part **11** Supporting part **12** Function-providing part **20** Power unit **25** First space **100** Aircraft **110** Propeller **112** First through-space **120** Holding part

Claims

1. An aircraft, comprising: a power unit having a first space open at least on one of an upper surface or a lower surface of the power unit; a propeller having a first through-space and connected to the power unit; a pair of holding parts defined by an arm or a frame of the aircraft and holding the power unit and the propeller on one side of the pair of holding parts; and a functional part having a predetermined function and including a grounding part that is configured to be in contact with a ground when landed, wherein the grounding part is coaxially arranged with the power unit and is located at least in part within an internal space formed by the first space and the first through-space, and the grounding part has an end protruding from the first space of the power unit and mounted to the one side of the pair of holding parts.
2. The aircraft according to claim 1, wherein the functional part has a non-rotating configuration.
3. The aircraft according to claim 1, wherein a portion of the functional part protrudes from the first through-space of the propeller.
4. The aircraft according to claim 1, wherein the first space is a second through-space through the power unit.
5. The aircraft according to claim 1, wherein the functional part comprises: a support part at least partially located in the internal space; and a function-providing part connected to the support part and providing the predetermined function.
6. The aircraft according to claim 5, wherein the aircraft includes a propeller guard as the function-providing part.
7. The aircraft according to claim 2, wherein a portion of the functional part protrudes from the first through-space of the propeller.
8. The aircraft according to claim 2, wherein the first space is a second through-space through the power unit.
9. The aircraft according to claim 3, wherein the first space is a second through-space through the power unit.
10. The aircraft according to claim 7, wherein the first space is a second through-space through the power unit.
11. The aircraft according to claim 3, wherein the portion of the functional part is mounted to the pair of holding parts.
12. The aircraft according to claim 11, wherein the propeller is mounted to a lower the one side of the holding part via the functional part.
13. The aircraft according to claim 1, wherein the grounding part is cylindrical with a center axis.
14. The aircraft according to claim 13, wherein the power unit is rotatable about the center axis of the grounding part.
15. The aircraft according to claim 1, further comprising a further power unit and a further propeller, wherein the pair of holding parts further holds the further power unit and the further propeller on the one side of the pair of holding parts.

16. The aircraft according to claim 15, wherein the one side of the pair of holding parts is a lower side of the pair of holding parts and is configured to face the ground when landed.
